

BLOOMBERG PORT

MAC 3 MULTI-ASSET CLASS RISK MODEL

A Complete Technical Deep Dive — From First Principles to Practice

Cross-Sectional Regression · Factor Quality Criteria · USE4 Methodology
Eigenfactor Risk Adjustment · Bayesian Shrinkage · DTS for High-Yield

Primary methodology reference:

Menchero, Orr & Wang (2011) — The Barra US Equity Model (USE4) Methodology Notes
Ben Dor, Dynkin & Hyman (2007) — Duration Times Spread

2025 Edition

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1. Introduction: Why a Multi-Asset Risk Model?

Modern portfolios hold equities, government bonds, corporate credit, foreign-currency assets, and commodities simultaneously. These assets do not move independently: when equity markets fall sharply, credit spreads widen, government bonds often rally, and currencies shift. A risk framework that treats each asset class in isolation gives a deeply misleading picture of total portfolio risk.

The **Bloomberg PORT MAC 3 (Multi-Asset Class 3)** model is Bloomberg's unified answer: a single factor model spanning all major asset classes, decomposing portfolio risk into interpretable economic factors.

1.1 The Dimensionality Problem

The most direct route to portfolio risk is the full $N \times N$ asset covariance matrix Σ , giving portfolio variance $\sigma_p^2 = \mathbf{w}'\Sigma\mathbf{w}$. The problem: Σ has $N(N+1)/2$ free parameters, none of which can be reliably estimated from limited history.

Portfolio size (N)	Parameters in Σ	Years of daily data needed
50 assets	1,275	~5 years
500 assets	125,250	~500 years
2,000 assets	2,001,000	> 7,000 years
10,000 assets	50,005,000	Impossible

Table 1.1: Direct covariance estimation rapidly becomes infeasible.

As Menchero, Orr & Wang (2011) articulate: "If the number of time observations is less than the number of stocks, the matrix is said to be rank deficient, meaning that it is possible to construct apparently riskless portfolios." Factor models solve this by reducing parameters to $O(K^2) + N$, where K (≈ 50 – 100) is far smaller than N .

1.2 Bloomberg PORT and MAC 3

- Covers equities, government bonds, IG and HY corporate credit, agency MBS, currencies, and commodities in one unified framework.
- Uses **observable fundamental characteristics** (market cap, duration, spread, credit rating) as factor exposures — not statistical constructs.
- Estimates factor returns via **cross-sectional regression** at each time step.
- Incorporates **Duration Times Spread (DTS)** for high-yield bond credit risk.
- Provides 700+ equity, 1,000+ fixed income, 300+ commodity, and 340+ currency factors.

1.3 José Menchero and the Intellectual Lineage

José Menchero (Head of Portfolio Analytics Research, Bloomberg LP) is the primary architect of MAC 3. Before Bloomberg, he spent eight years at **MSCI Barra** as Managing Director, responsible for

developing the Barra equity risk model family including **GEM2**, **GEM3**, and the US equity model **USE4**. Co-authors of the USE4 methodology notes — **D.J. Orr** and **Jun Wang** — were also Bloomberg colleagues.

Intellectual lineage: Barra E3 (1990s) → GEM2 (Menchero, Morozov & Shepard, 2008) → **USE4 (Menchero, Orr & Wang, 2011)** → Bloomberg MAC 1, MAC 2 → **MAC 3** (current). Each generation extended coverage and sharpened estimation methodology. The USE4 paper is the most complete public exposition of the methodology underlying MAC 3's equity model.

2. Linear Factor Model: First Principles

2.1 The Fundamental Equation (USE4 Eq. 1.1–1.5)

Following Menchero, Orr & Wang (2011), the excess return of asset n in period t is attributed to K common factors plus an idiosyncratic residual:

The Factor Model Equation (USE4 Eq. 1.1):

$$r_{n,t} = \sum_k X_{nk} f_{k,t} + u_{n,t}$$

Matrix form (N assets, K factors): $\mathbf{r}_t = \mathbf{X}_t \mathbf{f}_t + \mathbf{u}_t$

Symbol	Dimension	What it is	Who knows it?
$\mathbf{r}_t$	$N \times 1$	Asset excess returns	Observable market data
$\mathbf{X}_t$	$N \times K$	Factor exposure matrix	Pre-computed fundamentals
$\mathbf{f}_t$	$K \times 1$	Factor returns this period	Unknown — estimated by CSR
$\mathbf{u}_{n,t}$	$N \times 1$	Specific (idiosyncratic) returns	Unknown residual

Table 2.1: Four components of the factor model (USE4 notation).

Two structural assumptions (USE4 p.6) make the model tractable:

- **Factors and specific returns are uncorrelated:** $E[f_k \cdot u_n] = 0$. Common shocks and idiosyncratic shocks are orthogonal.
- **Specific returns are mutually uncorrelated:** $E[u_n \cdot u_m] = 0$ for $n \neq m$. Each stock's idiosyncratic term is its own, not shared with any other stock.

2.2 Portfolio Variance Decomposition (USE4 Eq. 1.2–1.5)

For a portfolio with weight vector $\mathbf{w}$ (USE4 Eq. 1.2), the portfolio return is $R_P = \sum_n w_n r_n$. The portfolio factor exposures (USE4 Eq. 1.3) are:

$$X_k^P = \sum_n w_n X_{nk}$$

Taking variances and using the two assumptions above (USE4 Eq. 1.5):

$$\begin{aligned} \text{Var}(R_P) &= (\mathbf{w}' \mathbf{X}) \mathbf{F} (\mathbf{X}' \mathbf{w}) + \mathbf{w}' \Delta \mathbf{w} \\ &= [\text{Systematic variance}] + [\text{Specific variance}] \end{aligned}$$

where:

$$\mathbf{F} = \text{Cov}(\mathbf{f}_t) \quad K \times K \text{ factor covariance matrix}$$

$$\Delta = \text{diag}(\sigma^2_{u,n}) \quad N \times N \text{ diagonal specific variance matrix}$$

Full asset covariance: $\Sigma = \mathbf{X} \mathbf{F} \mathbf{X}' + \Delta$

Why this matters: The $N \times N$ matrix Σ is never computed directly. All risk calculations operate on the compact triplet (X, F, Δ) . With $K = 60$ factors, the factor covariance matrix has fewer than 2,000 independent elements — a 99.8% reduction from the 2-million parameters needed for a 2,000-stock universe.

2.3 Intuition: Why Factor Models Work

Consider Apple (large-cap tech, high momentum) and Goldman Sachs (large-cap financials, low momentum). Their correlation arises almost entirely from *shared factor exposures* — both have equity market beta near 1.2. Their divergence comes from differing sector and style exposures. The factor model quantifies this exactly:

Risk source	Apple	Goldman Sachs	Nature in model
Market factor	$\beta = 1.15$	$\beta = 1.20$	Shared — drives correlation
Tech sector	+1.8	≈ 0	Divergent
Financials sector	≈ 0	+1.6	Divergent
Momentum	+1.3	-0.4	Divergent
Specific return	u_{AAPL}	u_{GS}	Uncorrelated by assumption

Table 2.2: Factor structure decomposing the Apple–Goldman covariance.

3. Cross-Sectional Regression: The Core Engine

We know X_t (factor exposures — observable fundamentals) and observe r_t (asset returns). The unknown is f_t : the factor returns this period. The cross-sectional regression (CSR) estimates them by treating the assets as observations and the factors as coefficients.

3.1 Setup and Notation

At each period t , minimise the weighted sum of squared residuals across all N assets:

Minimise over f_t :

$$Q(f_t) = (r_t - X_t f_t)' W_t (r_t - X_t f_t)$$

Solution (Weighted Least Squares):

$$\hat{f}_t = (X_t' W_t X_t)^{-1} X_t' W_t r_t$$

W_t = diagonal weight matrix

(market-cap weights initially; $1/\sigma_{u,n}^2$ after iteration)

3.2 OLS Worked Example — Full Step-by-Step

Step 0: The data for one period

Six stocks, three factors: **Market** (all stocks have exposure 1.0), **Size** (standardised log-market-cap), and **Value** (standardised book-to-price).

Stock	Return r_i (%)	Market (x_M)	Size (x_S)	Value (x_V)
Alpha Corp	+2.80	1.0	+1.32	-0.54
Beta Inc	+0.40	1.0	+0.21	+1.10
Gamma Ltd	+3.50	1.0	+0.78	+0.93
Delta PLC	-1.20	1.0	-1.48	-0.31
Epsilon SA	+1.90	1.0	-0.15	-0.88
Zeta GmbH	-0.60	1.0	+0.82	+0.44

Table 3.1: Cross-sectional data. Market exposure = 1.0 for all equities.

Step 1–4: Matrix computation

$X'X$ (3x3 moment matrix):

Market row: [6.00, 1.50, 0.74]

Size row: [1.50, 5.28, 1.20]

Value row: [0.74, 1.20, 3.43]

$X'r$ (right-hand side vector):

Market: sum of all returns = 6.80
 Size: sum of ($x_{S,i} * r_i$) = 7.51
 Value: sum of ($x_{V,i} * r_i$) = 0.62

Step 5: Solve $\hat{f} = (X'X)^{-1} X'r$

Estimated factor returns for this period:

$\hat{f}_{\text{Market}} = +0.87\%$ $\hat{f}_{\text{Size}} = +1.28\%$ $\hat{f}_{\text{Value}} = -0.45\%$

Step 6: Specific (idiosyncratic) returns

Stock	r_i (%)	Factor contrib. (%)	Specific u_i (%)
Alpha Corp	+2.80	$0.87+1.69+0.24 = +2.80$	≈ 0.00
Beta Inc	+0.40	$0.87+0.27-0.50 = +0.64$	-0.24
Gamma Ltd	+3.50	$0.87+1.00-0.42 = +1.45$	+2.05
Delta PLC	-1.20	$0.87-1.89+0.14 = -0.88$	-0.32
Epsilon SA	+1.90	$0.87-0.19+0.40 = +1.08$	+0.82
Zeta GmbH	-0.60	$0.87+1.05-0.20 = +1.72$	-2.32

Table 3.2: Gamma Ltd's +2.05% and Zeta GmbH's -2.32% are pure stock-specific events — not explained by any common factor.

3.3 Country Factor and Collinearity Constraint (USE4 Eq. 3.2–3.5)

In a global equity model, adding a Country factor introduces an exact collinearity: the sum of a stock's industry factor exposures always equals 1.0 (its country exposure). USE4 Equation 3.2 states this formally:

$$\sum_i X_{n, \text{industry}_i} = 1 \quad \text{for all stocks } n$$

Without a constraint, the regression is undefined — infinitely many (country, industry) combinations fit equally well. **Menchero, Orr & Wang (2011)** resolve this with a constraint that simultaneously removes collinearity and gives a clean economic interpretation to each factor:

USE4 Constraint (Eq. 3.3):

$$\sum_i w_i f_{i,t} = 0$$

The **cap-weighted sum of industry factor returns equals zero**. The overall market return is entirely absorbed by the Country factor.

Consequence (USE4 Eq. 3.4–3.5): the return of the cap-weighted estimation universe reduces to:

$$R_E \sim f_{\text{country}}$$

The Country factor return essentially equals the cap-weighted universe return (USE4 time-series correlation: 99.9%).

Industry factor $i \Rightarrow$ dollar-neutral long industry / short market.

Style factor $k \Rightarrow$ unit exposure to style k , zero to all others.

Why this constraint matters for practitioners: Without it, a 'Tech sector' factor return of +2% might be partly market, partly pure tech. With the constraint, the Country factor absorbs the market, and the Tech factor return represents *pure tech outperformance relative to the market*. Risk attribution becomes clean and non-overlapping.

3.4 Factor Portfolio Interpretation

As shown by Menchero (2010), each factor return can be interpreted as the return to a **pure factor portfolio**:

Factor	Portfolio interpretation	Economic meaning
Country (Market)	Cap-weighted estimation universe	Overall market return
Industry i	100% long industry i , 100% short market, zero style exposure	Pure industry return net of market and all style effects
Style k	Unit exposure to style k , zero industry exposure, dollar-neutral	Pure style premium: e.g., pure value vs growth return

Table 3.3: Factor portfolio interpretation (Menchero 2010, USE4 §3.1).

3.5 GLS and Iterated WLS

OLS assumes equal residual variance across all stocks — clearly wrong. GLS weights by inverse specific variance:

GLS: $\hat{f} = (X' \Delta^{-1} X)^{-1} X' \Delta^{-1} r$

$\Delta = \text{diag}(\sigma_u^2, 1, \dots, \sigma_u^2, N)$

MAC 3 iterates to true GLS:

1. Initialise W = market-cap weights.
2. Estimate $\hat{f} = (X'WX)^{-1} X'Wr$.
3. Compute residuals $\hat{u}_n = r_n - x_n' \hat{f}$.
4. Update σ_u^2, n from $\hat{u}_n^2$ (Huber M-estimator).
5. Set $W = 1/\sigma_u^2, n$. Return to step 2 until convergence.

Why iterate? The specific variances needed for GLS are themselves unknown before running the regression. Menchero's iterative approach jointly estimates factor returns and specific variances, converging to a consistent solution that delivers 15–25% lower estimation error vs. simple cap-weighted OLS.

4. CSR vs Time-Series Regression vs PCA

Three paradigms exist for building a factor risk model. Understanding why MAC 3 chose cross-sectional regression requires understanding what the alternatives cannot do.

4.1 Time-Series Regression (Fama-French Style)

For each asset n , regress over $t = 1..T$ historical periods:

$$r_{n,t} = \alpha_n + \beta_{n,M} \text{MKT}_t + \beta_{n,S} \text{SMB}_t + \beta_{n,V} \text{HML}_t + \epsilon_{n,t}$$

Result: N separate regressions. Factor returns (MKT, SMB, HML) are pre-defined traded portfolios. Factor exposures (betas) are estimated.

- **Requires 36–60 months of history per asset.** New IPOs or bonds have no history.
- **Stale exposures.** A spin-off overnight changes a stock's true profile, but TSR betas reflect the old company for years.
- **Does not extend to multi-asset.** Equity and bond betas live in incompatible spaces.

4.2 PCA / Statistical Factor Models

PCA extracts factors and exposures purely from return data with no economic priors.

- **Interpretability is zero.** No portfolio manager can intuit 'PC3'.
- **Unstable.** Eigenvectors can flip sign between estimation windows.
- **Cannot handle new assets.** A new bond with no return history gets no loading.

4.3 Fundamental CSR: MAC 3's Choice

- **Factor exposures are observable today.** A new bond issued yesterday gets DTS exposure immediately from its duration and spread.
- **Factor returns are estimated from the cross-section.** 2,000 stocks' returns today estimate 3 factor returns — not 60 months of one stock's history.
- **Economically meaningful.** 'Size returned +1.28% today' vs. 'PC3 had eigenvalue 0.003'.
- **Multi-asset extension is natural.** Equities get equity exposures; bonds get fixed income exposures. Both live in the same regression.

Criterion	Time-Series Regression	PCA / Statistical	Fundamental CSR (MAC 3)
Factor exposures	Estimated from history	From eigenvectors	Observable fundamentals
Factor returns	Pre-defined portfolios	Linear combos of r	Estimated by CSR
New assets	Cannot handle	Cannot handle	Immediate
Interpretability	Moderate (betas)	Poor	Excellent
Stability	Moderate	Low	High

Criterion	Time-Series Regression	PCA / Statistical	Fundamental CSR (MAC 3)
Multi-asset	Awkward	Very difficult	Natural
Used in MAC 3?	No	Diagnostics only	YES — core

Table 4.1: Three paradigms compared. MAC 3 uses fundamental CSR as its core engine.

5. The MAC 3 Factor Taxonomy

MAC 3 covers over 2,000 factors across asset classes. The structure below follows the Menchero lineage: country / industry / style for equities, analytical sensitivities for fixed income, and commodity term-structure factors.

5.1 Equity Factors (700+)

Factor class	Examples	Exposure type
World / Market	Single global market beta	1.0 for all equities (USE4 Country factor)
Country	US, UK, Japan, Brazil, ...	Binary (1 for home country) + beta
Industry / GICS	Technology, Financials, Healthcare, ...	Binary or multiple-industry fraction (USE4 §2.4–2.5)
Size	Log market capitalisation	Standardised: $(\ln(\text{mktcap}) - \mu_{cw}) / \sigma_{ew}$
Value	Book-to-price, earnings yield, ...	Composite of descriptors, re-standardised
Momentum	12-month minus 1-month return	Standardised descriptor
Volatility	Historical return volatility	Standardised descriptor
Growth	Earnings/sales growth forecasts	Composite descriptor
Leverage	Debt-to-equity, interest coverage	Standardised descriptor

Table 5.1: Equity factor categories in MAC 3.

5.2 Fixed Income Factors (1,000+)

Factor class	Exposure	Assets
Parallel shift (DV01)	Modified Duration	All bonds
Key Rate Durations	KRD 2y, 5y, 10y, 30y	Govt + IG corp
Credit Spread Duration	Spread Duration	IG corporate
DTS (HY credit)	Duration × Spread	HY corporate
Prepayment	Option-adjusted duration	Agency MBS
Currency	FX beta to each currency pair	Non-base ccy bonds

Table 5.2: Fixed income factor exposures in MAC 3.

5.3 Commodity and Currency Factors

- **Commodities (300+):** Constant-maturity futures exposures across energy, metals, agriculture, and livestock. Separate term-structure factors (front-month vs. back-month) and volatility-aware structures.

- **Currencies (340+):** FX factor exposures to G10 and EM currency pairs, carry and momentum factors, and a country-residual factor.

5.4 Cross-Asset Linkages

Equity–bond linkage: MAC 3 captures the empirical negative correlation between equity returns and government bond returns (“flight to quality”) through a shared market regime factor. A portfolio short equities and long Treasuries gets credit for this diversification in MAC 3, whereas asset-class-specific models would miss it entirely.

6. Factor Exposure Quality Criteria (USE4 §2)

Sound factor structure is a key pillar of a high-quality risk model. Menchero, Orr & Wang (2011, §2.1) enumerate five criteria that every factor must satisfy before entering MAC 3: **significance**, **stability**, **parsimony**, **low collinearity**, and **intuitive interpretability**.

6.1 Factor Stability Coefficient (USE4 Eq. 2.1)

Stability measures how much factor exposures change month-to-month. Drastic changes are problematic for risk management because they create phantom risk — apparent risk from factor rotation rather than true portfolio change. The stability coefficient is defined as (USE4 Eq. 2.1):

$$\begin{aligned} \rho_k(t) &= \text{Cross-sectional correlation of exposures from } t-1 \text{ to } t \\ &= \text{Cov}_w(X_{nk}^t, X_{nk}^{(t-1)}) / (\text{std}_w(X_{nk}^t) * \text{std}_w(X_{nk}^{(t-1)})) \end{aligned}$$

where w_n are regression weights (market-cap based).

Rule of thumb (USE4 p.7):

$\rho_k > 0.90 \Rightarrow$ Acceptable stability
 $\rho_k < 0.80 \Rightarrow$ Too unstable for model inclusion

Factor	Typical stability ρ_k	Verdict
Market beta	> 0.97	Very stable — beta changes slowly
Size (log mktcap)	> 0.99	Extremely stable — market cap moves slowly
Momentum	0.88–0.94	Moderate — 12m return window smooths changes
Short-term reversal	0.40–0.60	Unstable — 1-month return flips constantly
Industry (GICS)	> 0.99	Very stable — sector rarely changes

Table 6.1: Illustrative stability coefficients. Short-term reversal fails the threshold and is excluded from MAC 3's exposure matrix.

6.2 Variance Inflation Factor — Collinearity (USE4 Eq. 2.2–2.3)

Collinearity occurs when one factor can be approximately replicated by a linear combination of others. Excessive collinearity inflates regression standard errors, making factor returns noisy and hard to interpret. The **Variance Inflation Factor (VIF)** detects this by regressing each factor against all others (USE4 Eq. 2.2):

Regress factor k 's exposures on all other factors:

$$X_{nk} = \sum_{j \neq k} b_j * X_{nj} + \text{delta}_{nk}$$

$$\text{VIF}_k = 1 / (1 - R^2_k) \quad [\text{USE4 Eq. 2.3}]$$

where R^2_k is the R-squared of the regression above.

$VIF_k = 1 \Rightarrow$ No collinearity (factor k is orthogonal to all others)

$VIF_k = 5 \Rightarrow$ Moderate collinearity ($R^2 = 80\%$)

$VIF_k > 10 \Rightarrow$ Severe collinearity $\Rightarrow$ exclude or orthogonalise

Orthogonalisation (Menchero 2010): When VIF is high, a factor can be “cleaned” by subtracting its projection onto the other factors. For example, the Non-Linear Size factor (quadratic in log-mktcap) is highly correlated with the linear Size factor. After orthogonalisation, Non-Linear Size captures the mid-cap vs. overall-market effect — intuitively, it becomes ‘mid-cap premium’.

6.3 Outlier Treatment: Three-Group Algorithm (USE4 §2.2)

Before computing exposures, raw descriptors are screened for outliers. USE4 uses a **three-group algorithm**:

Group	Condition	Action	Rationale
Group 1 — Extreme	$ x - \mu > 4\sigma$	Remove from estimation	Likely data error; prevents single outlier from dominating regression
Group 2 — Large	$3\sigma < x - \mu \leq 4\sigma$	Winsorise to $\mu \pm 3\sigma$	Legitimate but extreme; cap its influence
Group 3 — Normal	$ x - \mu \leq 3\sigma$	Use as-is	Clean data; no adjustment needed

Table 6.2: Three-group outlier treatment (USE4 §2.2). Applied at the descriptor level, not the stock return level.

6.4 Style Exposure Standardisation (USE4 Eq. 2.4–2.5)

Raw descriptors vary widely in scale (log-market-cap ranges 0–15; book-to-price ranges 0–5). Standardisation puts all exposures on a common scale. USE4 Equation 2.4 defines the standardised descriptor:

$$d_{nl} = (Raw_{nl} - \mu_l) / \sigma_l \quad [USE4 \text{ Eq. 2.4}]$$

where:

μ_l = cap-weighted cross-sectional mean of descriptor l

σ_l = equal-weighted cross-sectional standard deviation of descriptor l

KEY DESIGN CHOICE (USE4 p.9):

μ uses cap-weights $\Rightarrow$ well-diversified cap-weighted portfolio has zero exposure

σ uses equal-weights $\Rightarrow$ large-caps do not dominate the scale

Multiple descriptors within one style factor are then combined (USE4 Eq. 2.5):

$$X_{nk} = \sum_l w_l * d_{nl} \quad [USE4 \text{ Eq. 2.5}]$$

where w_l are descriptor weights optimised to maximise model explanatory power. The final style factor exposure is then re-standardised to:
cap-weighted mean = 0, equal-weighted std = 1.

6.5 Multiple-Industry Exposures (USE4 Eq. 2.6–2.8)

Many companies have material revenue across several industries. A simple binary industry assignment (“IBM is Technology”) misses this. USE4 computes fractional industry exposures using two signals (USE4 §2.5):

Assets-based: regress mkt cap against assets per industry [USE4 Eq. 2.6]

$$M_n = \sum_k (\alpha_k * A_{nk}) + \delta_n$$

$$X_{nk}^A = \alpha_k * A_{nk} / M_n \quad (\text{fraction of value from industry } k)$$

[USE4 Eq. 2.7]

Sales-based: same regression with Sales instead of Assets.

Combined exposure (USE4 Eq. 2.8):

$$X_{nk} = 0.75 * X_{nk}^{\text{Assets}} + 0.25 * X_{nk}^{\text{Sales}}$$

Constraint: $\sum_k X_{nk} = 1$ (exposures sum to 1 per stock).

Maximum 5 industry exposures per stock; re-normalised if needed.

Why this matters: Amazon has revenue from e-commerce (Consumer Discretionary), cloud services (Technology), and advertising (Communication Services). A binary classification misses the Technology and Communication components entirely, understating its correlation with tech stocks. MAC 3’s fractional exposures capture this multi-business reality.

6.6 Worked Example: Size Exposure Calculation

Stock	Mkt Cap (\$B)	ln(Mkt Cap)	Wtd mean (μ)	Wtd std (σ)	Z-score (x_s)
MegaCorp	450.0	6.109	4.510	1.284	+1.25
BigCo	85.0	4.443	4.510	1.284	-0.05
MidCo	22.0	3.091	4.510	1.284	-1.10
SmallCo	4.5	1.504	4.510	1.284	-2.34
Micro Inc	0.8	-0.223	4.510	1.284	-3.69 → clipped to -3.00

Table 6.3: Size factor exposure via log-mktcap standardisation. μ uses cap-weights; σ uses equal-weights. Micro Inc is winsorised at 3σ .

If Size factor return is estimated at +0.87% in a given period, MegaCorp earns $1.25 \times 0.87\% = +1.09\%$ from the Size factor, while SmallCo earns $-2.34 \times 0.87\% = -2.04\%$.

7. Fixed Income Factor Exposures and DTS

Fixed income factor exposures are **analytical sensitivities**, derived from the bond's cash flows and current market prices — not statistical regressions. They are available immediately for any new bond issued today.

7.1 Analytical Sensitivities vs Statistical Betas

Exposure	Definition	Formula	Used for
Modified Duration	Price sensitivity to parallel yield shift	$-(1/P) dP/dy$	Interest rate risk: all bonds
Key Rate Duration (KRD)	Price sensitivity to shift at maturity point k	$-(1/P) dP/dy_k$	Curve shape risk: govt, IG corp
Spread Duration	Price sensitivity to credit spread shift	$-(1/P) dP/ds \approx \text{ModDur}$	IG credit spread risk
DTS	Spread duration scaled by current OAS	$\text{ModDur} \times \text{OAS}$	HY credit spread risk (see below)

Table 7.1: Fixed income factor exposures in MAC 3.

7.2 Duration Times Spread (DTS) — Derivation from First Principles

The DTS concept (Ben Dor, Dynkin & Hyman, 2007) arises from a key empirical regularity: for high-yield bonds, **spread changes are proportional to the level of the spread**, not additive. A bond trading at 800 bps typically moves 80 bps when spreads “widen 10%”; a bond at 200 bps moves only 20 bps.

Step 1: Excess return from spread change

$$XR \approx -\text{ModDur} \times \Delta s$$

where Δs = change in OAS in decimal form. This is the standard duration approximation applied to credit spreads.

Step 2: Spread volatility scales with spread level

The empirical regularity for HY bonds (Ben Dor et al., 2007):

$$\sigma(\Delta s) \approx \kappa \times s$$

where $\kappa \approx 0.40\text{--}0.50$ for HY bonds and s is the current OAS level (in decimal). Spreads are *multiplicatively* (not additively) mean-reverting.

Step 3: Excess return volatility

```
sigma(XR) = ModDur * sigma(Delta_s)
           = ModDur * kappa * s
           = kappa * (ModDur * s)
           = kappa * DTS
```

where $DTS = \text{Modified Duration} \times OAS \text{ (in decimal)}$

Therefore $\sigma(XR)$ is proportional to DTS, not to spread duration alone.

This is why HY bonds must use DTS as their credit risk exposure, not SpreadDur.

Why IG bonds use Spread Duration but HY uses DTS:

For IG bonds (OAS 50–200 bps), spread changes are roughly additive: a 10 bps widening at 100 bps is similar to a 10 bps widening at 80 bps. Spread Duration works.

For HY bonds (OAS 300–1500+ bps), spread changes are multiplicative: a 10% widening at 800 bps = 80 bps widening, vs. 10% at 200 bps = 20 bps. A bond with twice the spread has roughly twice the spread volatility *and* twice the excess return volatility. DTS captures this: doubling spread doubles DTS.

7.3 DTS Worked Examples

Bond	Rating	Mod Dur	OAS (bps)	OAS (decimal)	DTS	$\sigma(XR) \approx$
IBM 2030	A	5.24y	85	0.0085	0.0446	$\kappa \times 0.0446$
Ford 2028	BB	3.72y	350	0.0350	0.130	$\kappa \times 0.130$
HCA 2030	B	5.10y	650	0.0650	0.332	$\kappa \times 0.332$
Sprint 2032	CCC	6.80y	1200	0.1200	0.816	$\kappa \times 0.816$

Table 7.2: DTS calculation. $\kappa \approx 0.45$ for HY bonds. Note how Sprint has 18× higher DTS than IBM despite similar maturity — correctly reflecting its vastly higher spread risk.

7.4 Fixed Income Exposure Matrix Example

Bond	Maturity	OAS (bps)	ModDur	KRD 2y	KRD 5y	KRD 10y	Credit Exposure
US Tsy 2y	2y	0	1.87	1.87	0.00	0.00	N/A (govt)
US Tsy 10y	10y	0	8.50	0.00	0.00	8.50	N/A (govt)
IBM 2030 (A)	6y	85	5.24	0.00	2.10	3.14	SprdDur = 5.24y
Ford 2028 (BB)	4y	350	3.72	0.00	3.72	0.00	DTS = 0.130
HCA 2030 (B)	6y	650	5.10	0.00	2.00	3.10	DTS = 0.332

Table 7.3: Fixed income factor exposures. IG bonds use Spread Duration; HY bonds use DTS. KRDs decompose the rate risk along the curve.

8. Factor Covariance Matrix Construction (USE4 §4)

After running the cross-sectional regression for T historical periods, we have a T×K matrix of estimated factor returns. The goal is to estimate the K×K factor covariance matrix F from this history. USE4 (§4.1–4.3) details four innovations beyond simple EWMA: separate half-lives for volatilities and correlations, Newey-West correction, Eigenfactor Risk Adjustment, and Volatility Regime Adjustment.

8.1 EWMA Factor Return History

Each period's factor returns are retained. Older observations receive exponentially declining weight. For period τ (measured in trading days from today), the EWMA weight is proportional to:

$$\lambda^\tau \text{ where } \lambda = 2^{-1/H} \text{ and } H = \text{half-life in trading days}$$

After half-life H days, the weight is 50% of today's weight; after 2H days, 25%; after 3H, 12.5%, etc.

8.2 Separate Volatility and Correlation Half-Lives

A critical USE4 innovation (§4.1): factor volatilities and correlations are estimated with **different half-lives**. The covariance between factors i and j is assembled as (USE4 Eq. 4.1):

$$F_{\text{hat}_{ij}} = \sigma_i * \rho_{ij} * \sigma_j \quad [\text{USE4 Eq. 4.1}]$$

where σ_i = factor volatility (estimated with vol half-life H_v)
 ρ_{ij} = factor correlation (estimated with corr half-life H_c)
 $H_v \ll H_c$ (vol half-life much shorter than correlation half-life)

Why separate half-lives? (USE4 p.18) Sampling error on the diagonal (volatility) has little impact on matrix conditioning. But sampling error on the off-diagonal (correlation) strongly affects conditioning and can produce near-singular matrices. A longer correlation half-life reduces this error. Meanwhile, a shorter volatility half-life makes risk forecasts more responsive to current market conditions.

8.3 USE4 Calibrated Parameters

Model	Factor Vol Half-Life	NW Vol Lags	Factor Corr Half-Life	NW Corr Lags	VRA Half-Life
USE4S (Short-term)	84 days	5	504 days	2	42 days
USE4L (Long-term)	252 days	5	504 days	2	168 days

Table 8.1: USE4 factor covariance parameters (USE4 Table 4.1, p.19). NW = Newey-West. VRA = Volatility Regime Adjustment. Half-life of 84 days $\approx$ 4 calendar months; 504 days $\approx$ 2 years.

MAC 3 follows the same dual half-life architecture, applied globally across all asset classes. The short-term model (used for most risk management) has a factor volatility half-life of roughly 3–6 months and a correlation half-life of 18–24 months. The long-term model weights stability over responsiveness.

8.4 Newey-West Autocorrelation Correction

Monthly factor returns often exhibit serial autocorrelation: factor returns this month are somewhat predictable from last month. Ignoring this understates factor variance. The **Newey-West estimator** (with L lags) corrects for this:

```
sigma^2_k (NW) = EWMA_var(f_k) + 2 * sum_{l=1}^L w_l * EWMA_cov(f_k,t, f_k,t-l)
```

USE4S parameters: $L = 5$ lags for volatility, $L = 2$ lags for correlation.
The $w_l = 1 - 1/(L+1)$ are Bartlett kernel weights that downweight distant lags.

Impact: for momentum-driven factor returns with lag-1 autocorr ~ 0.20 , Newey-West increases the variance estimate by roughly 15-20%.

8.5 Eigenfactor Risk Adjustment

One of the key USE4 innovations (§4.2): when portfolios are **optimised** against the factor covariance matrix, the resulting risk forecasts are systematically *too low*. The reason: optimisation exploits the eigenvectors with the smallest eigenvalues (i.e., the “cheapest” directions in factor space), which are also the most subject to estimation error. The optimiser “tilts” the portfolio towards noisy low-eigenvalue directions that look artificially riskless.

USE4 result (Fig. 4.2): Using the unadjusted factor covariance matrix, 100 optimised portfolios were constructed. All 100 bias statistics fell *above* the 95% confidence interval — systematic underprediction of risk for optimised portfolios. After Eigenfactor Risk Adjustment (Fig. 4.5), most bias statistics fell within the confidence interval.

The procedure (USE4 Appendix B) uses **Monte Carlo simulation**:

1. Compute the $K \times K$ sample factor covariance matrix $\hat{F}$. Decompose into eigenvectors V and eigenvalues Λ : $\hat{F} = V\Lambda V'$.
2. Treating $\hat{F}$ as ‘truth’, simulate M histories of factor returns via Cholesky decomposition. Compute a simulated covariance matrix $\hat{F}_{\text{sim}}$ for each history.
3. Compare predicted eigenvalue volatility (from $\hat{F}_{\text{sim}}$) with ‘true’ eigenvalue (from $\hat{F}$). The simulated bias for eigenfactor k is: $\beta_k^{\text{sim}} = \lambda_k^{\text{sim}} / \lambda_k$.
4. Adjust each eigenvalue upward to correct for the simulated bias: $\Lambda_k^{\text{adj}} = \lambda_k \times (\lambda_k / \lambda_k^{\text{sim}})^\alpha$.
5. Rotate the adjusted eigenvalues back to the original factor basis: $\hat{F}^{\text{adj}} = V \Lambda^{\text{adj}} V'$.

Effect on eigenvalue spectrum:

```
Small eigenvalues (noisy directions) => adjusted UPWARD most
Large eigenvalues (well-estimated)  => adjusted upward least
```

Net effect: the optimiser can no longer exploit artificially low-risk directions.

8.6 Volatility Regime Adjustment (USE4 §4.3)

Factor volatilities estimated from a single EWMA are a backward-looking average. In a regime shift (e.g., the VIX spikes from 15 to 40), the EWMA takes weeks to catch up. The **Volatility Regime Adjustment (VRA)** calibrates factor volatilities to current market conditions by using the **cross-sectional dispersion of stock returns** as a real-time vol signal.

VRA methodology:

1. Compute cross-sectional standard deviation of stock returns each day: CSD_t
2. Estimate the long-run expected CSD via EWMA (VRA half-life H_{vra}):
$$CSD_{hat}_t = EWMA(CSD_t; H_{vra})$$
3. VRA multiplier: $\lambda_t = CSD_t / CSD_{hat}_t$
4. Adjust factor volatilities: $\sigma_k^{adj} = \lambda_t * \sigma_k^{EWMA}$

USE4S VRA half-life: 42 trading days (~2 months)

USE4L VRA half-life: 168 trading days (~8 months)

During market stress ($\lambda_t > 1$): forecasts increase rapidly.

During calm regimes ($\lambda_t < 1$): forecasts decrease, preventing overstatement.

9. Specific Risk Estimation (USE4 §5)

Specific (idiosyncratic) risk is the component of a stock's return variance not explained by the common factors. Even a perfectly estimated factor model has specific risk for every stock — it represents earnings surprises, management changes, litigation events, and other stock-specific news.

9.1 EWMA Specific Variance

The specific variance for stock n is estimated from the EWMA of squared specific returns $\hat{u}_{n,t}$ (the residuals from the CSR):

$$\sigma^2_{u,n} = \text{EWMA}(\hat{u}_{n,t}^2; H_{\text{spec}})$$

where H_{spec} = specific variance half-life (USE4: daily observations used, unlike monthly factor returns).

USE4 uses daily specific returns for higher-frequency specific risk estimates. This is blended with a structural model that uses fundamental data (Size, Volatility, Leverage) to provide a prior for stocks with short histories.

9.2 Bayesian Shrinkage Toward Size-Decile Mean (USE4 Eq. 5.6–5.8)

Without adjustment, EWMA specific variance estimates are biased: low-volatility stocks tend to have understated risk (their recent calm period may not persist), and high-volatility stocks tend to have overstated risk. USE4 Fig. 5.1 shows this clearly: the bias statistic rises monotonically from low- to high-volatility deciles without shrinkage.

The fix (USE4 §5.2): **Bayesian shrinkage** toward the cap-weighted mean specific volatility of the stock's size decile. The shrunk estimate $\hat{\sigma}_n^{\text{SH}}$ is (USE4 Eq. 5.6):

$$\hat{\sigma}_n^{\text{SH}} = v_n * \mu(s_n) + (1 - v_n) * \hat{\sigma}_n \quad [\text{USE4 Eq. 5.6}]$$

where:

$$\mu(s_n) = \text{cap-weighted mean specific vol for size decile } s_n \quad [\text{USE4 Eq. 5.7}]$$

$$= \sum_{m \in s_n} w_m * \hat{\sigma}_m$$

$$v_n = \text{shrinkage intensity (how much weight on the prior)} \quad [\text{USE4 Eq. 5.8}]$$

$$= q * \sigma(s_n) / (\hat{\sigma}_n + q * \sigma(s_n))$$

$$\sigma(s_n) = \text{cross-sectional std of specific vols in decile } s_n$$

$$q = \text{tuning parameter } (\sim 1.0 \text{ in USE4})$$

Interpretation of v_n : When there is high dispersion of specific vols within a size decile (large $\sigma(s_n)$), shrinkage is light (small v_n) because the individual estimates are informative. When all stocks in a decile have similar specific vols (small $\sigma(s_n)$), shrinkage is heavy because the

decile mean is almost as informative as the individual estimate.

Scenario	σ_n^*	$\mu(s_n)$	v_n	σ_n^{SH}	Effect
Low-vol stock, calm recent period	12%	18%	0.45	15.3%	Raised toward decile mean
High-vol stock, recent volatility spike	42%	22%	0.38	34.4%	Lowered toward decile mean
Typical stock	20%	19%	0.30	19.7%	Small adjustment

Table 9.1: Illustrative Bayesian shrinkage. $v_n \approx 0.30\text{--}0.45$ is typical. Both extremes are pulled toward the decile average.

10. Model Validation: Bias Statistics (USE4 Appendix A)

A bias statistic measures the ratio of *realised* portfolio risk to *predicted* portfolio risk. A well-calibrated model should produce bias statistics close to 1.0. Menchero, Orr & Wang (2011, Appendix A) provide the definitive formulation used across all Barra/MAC 3 models.

10.1 Single-Window Bias Statistic (USE4 Eq. A1–A3)

Let $R_{n,t}$ be the return to test portfolio n in period t , and $\sigma_{n,t}$ the beginning-of-period predicted volatility. The **standardised return** is (USE4 Eq. A1):

$$b_{n,t} = R_{n,t} / \sigma_{n,t}$$

Under perfect risk forecasts, $b_{n,t}$ has mean 0 and standard deviation 1. The **bias statistic** for portfolio n is the realised standard deviation of standardised returns (USE4 Eq. A2):

$$B_n = \sqrt{\frac{1}{(T-1)} \sum_t (b_{n,t} - \bar{b}_n)^2} \quad [\text{USE4 Eq. A2}]$$

where T = number of periods in the observation window,
 $\bar{b}_n$ = time-average of $b_{n,t}$.

$B_n = 1.0 \Rightarrow$ Perfect calibration

$B_n > 1.0 \Rightarrow$ Model underpredicts risk (realised vol > forecast)

$B_n < 1.0 \Rightarrow$ Model overpredicts risk (realised vol < forecast)

The 95% confidence interval for B_n under normality (USE4 Eq. A3):

Confidence interval (USE4 Eq. A3):

$$B_n \in [1 - \sqrt{(2/T)}, 1 + \sqrt{(2/T)}]$$

Examples: $T=60$ months $\rightarrow$ CI = [0.82, 1.18]. $T=120$ months $\rightarrow$ CI = [0.87, 1.13]. $T=191$ months (USE4 sample) $\rightarrow$ CI = [0.90, 1.10].

Non-normality caveat (USE4 p.37): For fat-tailed returns (kurtosis > 3), fewer than 95% of bias statistics fall within this interval even for perfect forecasts. At kurtosis = 5 (typical for equity returns), only ~86% fall in the CI for $T=120$. This means the CI must be interpreted conservatively — a bias statistic slightly above the upper bound is not necessarily evidence of model mis-calibration.

10.2 Rolling-Window Approach

Rather than one bias statistic over the full sample, USE4 uses a rolling window of T months advanced by one month at a time. This gives a time series of bias statistics that reveals *when* the model was well-calibrated vs. when it diverged:

Window	Advantage	Disadvantage	USE4 application
Full sample (T=191)	Low sampling error	Averages out periods of over/underprediction	Overall model summary statistic
Rolling 24-month	Reveals regime changes	High variance — many windows outside CI by chance	Factor-level calibration check
Rolling 60-month	Good balance	Lags in detecting persistent mis-calibration	Portfolio-level risk audit

Table 10.1: Single vs. rolling window bias statistics.

10.3 Applying Bias Statistics: Eigenfactor Adjustment Validation

USE4 demonstrates the bias statistic framework concretely by applying it to 100 optimised factor portfolios:

Covariance matrix used	Bias statistics result	Verdict
Unadjusted EWMA F^*	All 100 portfolios: $B_n \geq 1.1$ (above CI)	Systematic underprediction — model fails
After Eigenfactor Risk Adjustment	Most portfolios: B_n within [0.90, 1.10]	Well-calibrated — model passes
After Eigenfactor + VRA	Stable across vol regimes	Robust to market stress periods

Table 10.2: Bias statistic results from USE4 validation (USE4 Figs. 4.2, 4.5).

11. Portfolio Risk Attribution

11.1 Systematic vs Specific Risk

For a portfolio with weight vector $\mathbf{w}$ and portfolio factor exposures $\mathbf{b} = \mathbf{X}'\mathbf{w}$:

Portfolio variance = Systematic + Specific

$$\sigma_p^2 = \mathbf{b}' \mathbf{F} \mathbf{b} + \mathbf{w}' \Delta \mathbf{w}$$

$$\sigma_p = \sqrt{\mathbf{b}' \mathbf{F} \mathbf{b} + \mathbf{w}' \Delta \mathbf{w}}$$

Systematic: driven by shared factor movements (market, sector, style, credit).

Specific: driven by stock/bond-specific events uncorrelated across assets.

11.2 Marginal Contribution to Risk (MCTR)

The MCTR of asset n measures how much total portfolio volatility changes when the weight of asset n increases by 1%:

$$\text{MCTR}_n = \text{partial } \sigma_p / \text{partial } w_n$$

$$= (1/\sigma_p) * [\sum_k X_{nk} * (\mathbf{F} \mathbf{b})_k + \sigma_{u,n}^2 * w_n]$$

$$\text{Factor MCTR}_n: (1/\sigma_p) * \sum_k X_{nk} * (\mathbf{F} \mathbf{b})_k$$

$$\text{Specific MCTR}_n: (1/\sigma_p) * \sigma_{u,n}^2 * w_n$$

$$\text{Additive property: } \sum_n w_n * \text{MCTR}_n = \sigma_p$$

11.3 Fully Worked Multi-Asset Example

Asset	Weight	Type	Mkt Beta	Size x_s	HY DTS	Spec. σ (ann.)
S&P 500 ETF	30%	Equity	0.95	-0.10	—	4.5%
Apple Inc.	20%	Equity	1.25	+1.30	—	9.2%
US Tsy 10y	25%	Govt Bond	0.00	0	—	0.8%
Ford BB Bond	25%	HY Bond	0.00	0	0.130	6.1%

Table 11.1: Example balanced portfolio with equity and HY credit.

Portfolio factor betas $\mathbf{b} = \mathbf{X}'\mathbf{w}$:

Factor	b_k	Calculation
Equity Market	0.535	$0.30 \times 0.95 + 0.20 \times 1.25 = 0.285 + 0.250$
Size	+0.23	$0.30 \times (-0.10) + 0.20 \times (+1.30) = -0.03 + 0.26$
US Rates (10y)	2.13y	$0.25 \times 8.50y$

Factor	b_k	Calculation
HY DTS (BB)	0.033	0.25×0.130

Risk source	Contrib to σ_p (ann.)	% of total
Equity Market factor	3.8%	52%
Size factor	0.4%	6%
US Rates factor	1.1%	15%
HY DTS factor	0.7%	10%
Cross-factor (eq–bond)	–0.3%	–4% (diversification)
Specific risk (equities)	0.8%	11%
Specific risk (bonds)	0.6%	8%
Total portfolio σ	7.3%	100%

Table 11.2: Risk attribution for example portfolio.

Reading this attribution: Equity market risk dominates at 52%. Rate risk is meaningful (15%) and HY credit adds another 10%. The –4% cross-factor contribution reflects equity–bond diversification: rates tend to fall when equities sell off, cushioning the portfolio. This diversification benefit is invisible in single-asset-class risk systems but fully captured in MAC 3.

12. References

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